

Week 1 – Introduction to Programming: Getting Started with Python

■ Learning Objectives

- Understand what the Canadian Computing Competition (CCC) is and its format (5 problems, 3 hours).
- Install Python and a suitable IDE (Thonny, PyCharm, VS Code, or Replit).
- Write and run your first Python program.
- Use print statements and basic syntax to produce text and numbers.

■ Key Concepts

1. **Printing and Output:** Using `print()` to display text or numbers.
2. **Strings and Numbers:** Strings use quotes, numbers do not.
3. **Comments:** Use `#` for notes in your code.
4. **Variables:** Store values for later use.
5. **Basic Math:** `+`, `-`, `*`, `/`, `%`, `**` for arithmetic.

Example 1: Hello World and Simple Math

```
# This is your first Python program
print("Hello, World!")

# Printing multiple items
print("My name is Alice and I am learning Python.")

# Basic math operations
a = 5
b = 3
print("a + b =", a + b)
print("a * b =", a * b)
print("a / b =", a / b)
print("a % b =", a % b)
```

■ Practice Problems

Practice 1 – Print Your Own Message

Write a program that prints your name, grade, and favorite subject.

Hint: Use multiple `print()` statements.

Solution:

```
print("Name: David")
print("Grade: 9")
print("Favorite subject: Computer Science")
```

Practice 2 – Rectangle Area Calculator

Write a program that defines the width and height of a rectangle, computes its area, and prints the result.

Hint: Use variables and multiplication.

Solution:

```
width = 8
height = 5
area = width * height
print("The area of the rectangle is:", area)
```

Practice 3 – Personal Greeting

Ask the user for their name using input() and greet them by name.

Solution:

```
name = input("What is your name? ")
print("Nice to meet you,", name, "!")
```

■ More Examples and Tips

You can combine strings and variables using commas or formatted strings (f-strings).

```
name = "Alice"
age = 15
print("Name:", name, "| Age:", age)
# Using f-string for better formatting
print(f"My name is {name} and I am {age} years old.")
```

Common Syntax Errors

- Missing quotes around strings → `print(Hello)` ■
- Using wrong case in keywords → `Print()` ■
- Unclosed parentheses → `print('Hi'` ■

■ Homework Challenge

Write a program that asks for your first name, last name, and year of birth, then prints a message like this:

```
# Example Output
# Enter your first name: Alice
# Enter your last name: Smith
# Enter your birth year: 2010
# Hello Alice Smith, you are 14 years old in 2024!
```

Hint: You'll need to convert input from text to numbers using `int()`.

Expected Solution:

```
first = input("Enter your first name: ")
last = input("Enter your last name: ")
birth = int(input("Enter your birth year: "))
age = 2024 - birth
print(f"Hello {first} {last}, you are {age} years old in 2024!")
```